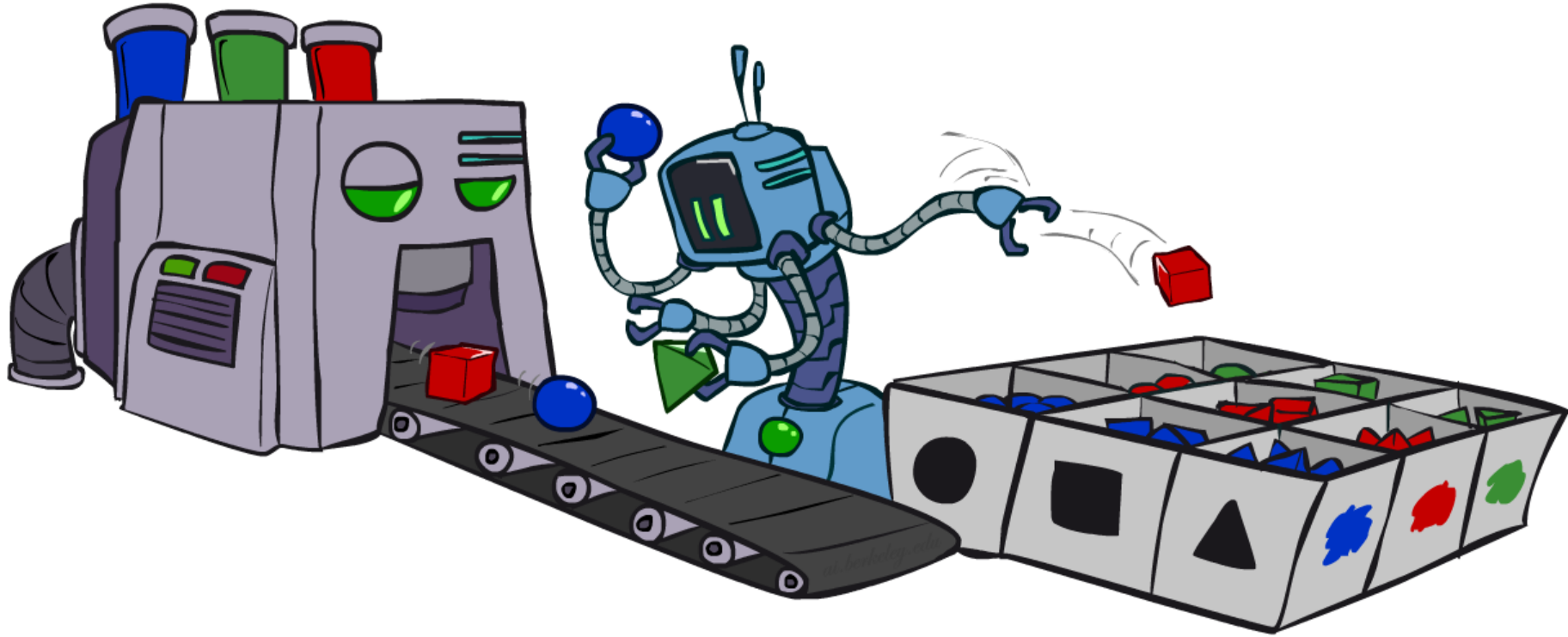


Bayes Nets: Approximate Inference



AIMA Chapter 14.5, PRML Chapter 11

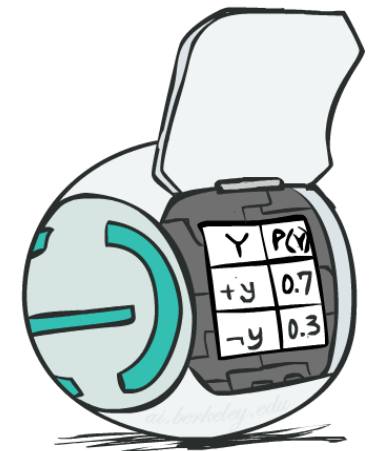
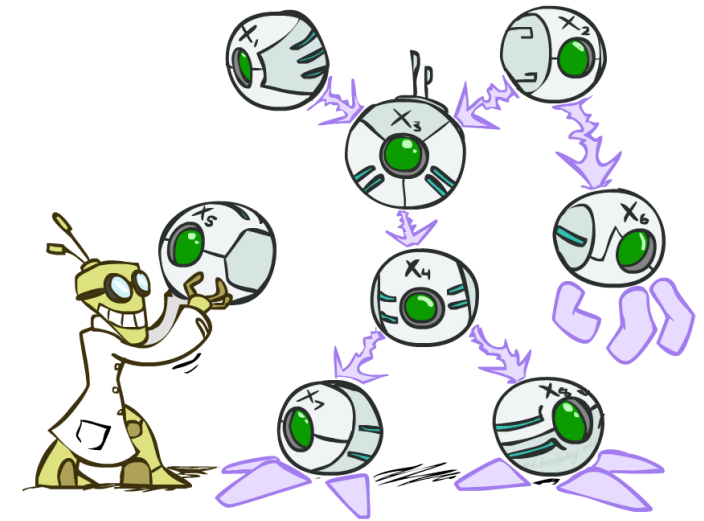
Bayes' Net Representation

- A directed, acyclic graph, one node per random variable
- A conditional probability table (CPT) for each node
 - A collection of distributions over X , one for each combination of parents' values

$$P(X|a_1 \dots a_n)$$

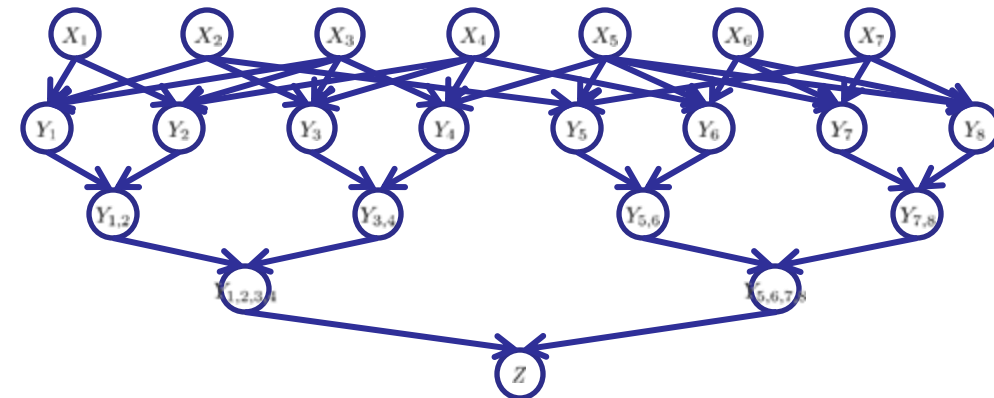
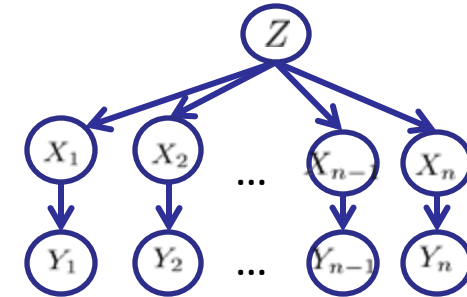
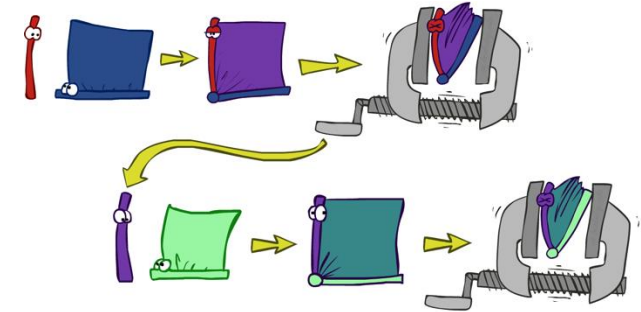
- Bayes' nets implicitly encode joint distributions
 - As a product of local conditional distributions
 - To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$



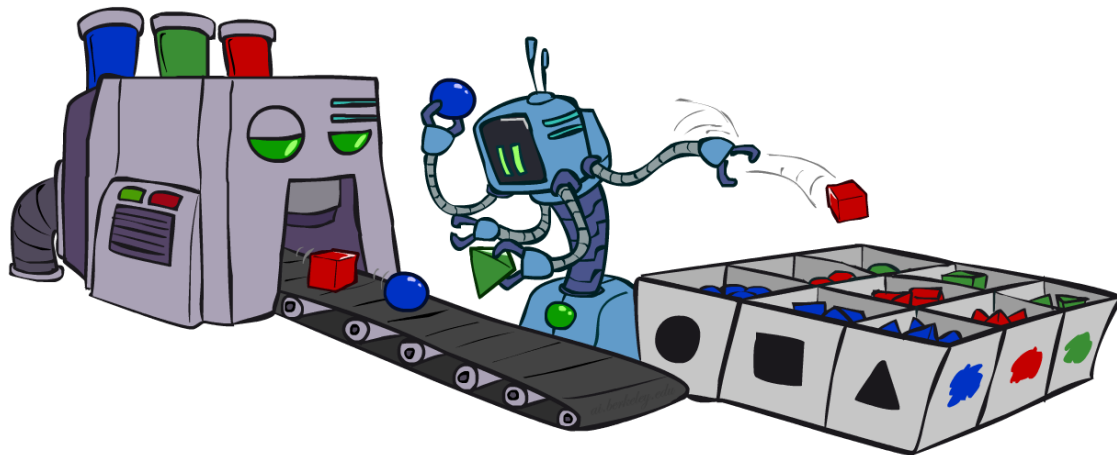
Variable Elimination

- Interleave joining and marginalizing
- d^k entries computed for a factor over k variables with domain sizes d
- Ordering of elimination of hidden variables can affect size of factors generated
- Worst case: running time exponential in the size of the Bayes' net



Sampling

- Goal: probability P
- Basic idea
 - Draw N samples from a sampling distribution S
 - Compute some quantity from the samples
 - Show this converges to the true probability P
- Why sample?
 - Often very fast to get a decent approximate answer
 - The algorithms are very simple and general (easy to apply to fancy models)
 - They require very little memory ($O(n)$)



Sampling from a discrete distribution

- Sampling from given distribution

- Step 1: Get sample u from uniform distribution over $[0, 1)$
 - Random() in many programming languages
- Step 2: Convert this sample u into an outcome for the given distribution by associating each outcome x with a $P(x)$ -sized sub-interval of $[0,1)$

- Example

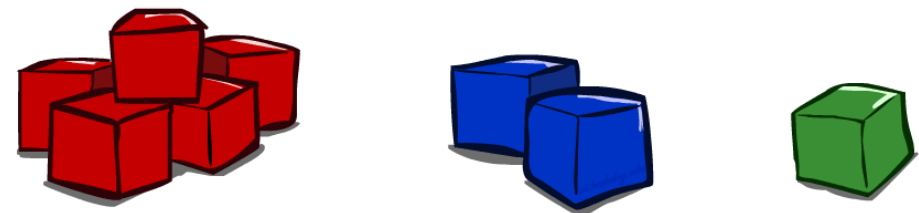
C	P(C)
red	0.6
green	0.1
blue	0.3

$$0 \leq u < 0.6, \rightarrow C = \text{red}$$

$$0.6 \leq u < 0.7, \rightarrow C = \text{green}$$

$$0.7 \leq u < 1, \rightarrow C = \text{blue}$$

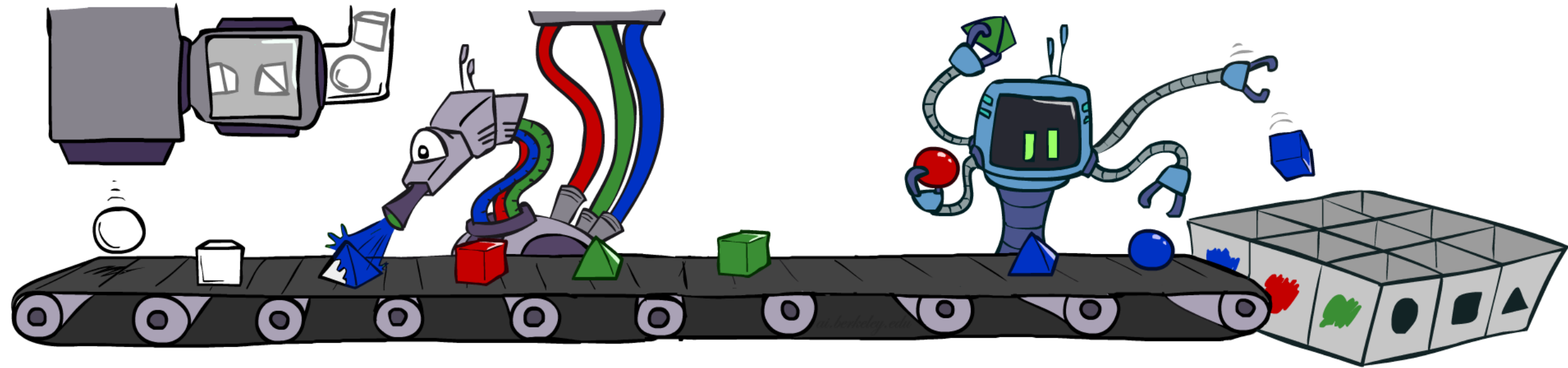
- If random() returns $u = 0.83$, then our sample is $C = \text{blue}$
- E.g, after sampling 8 times:



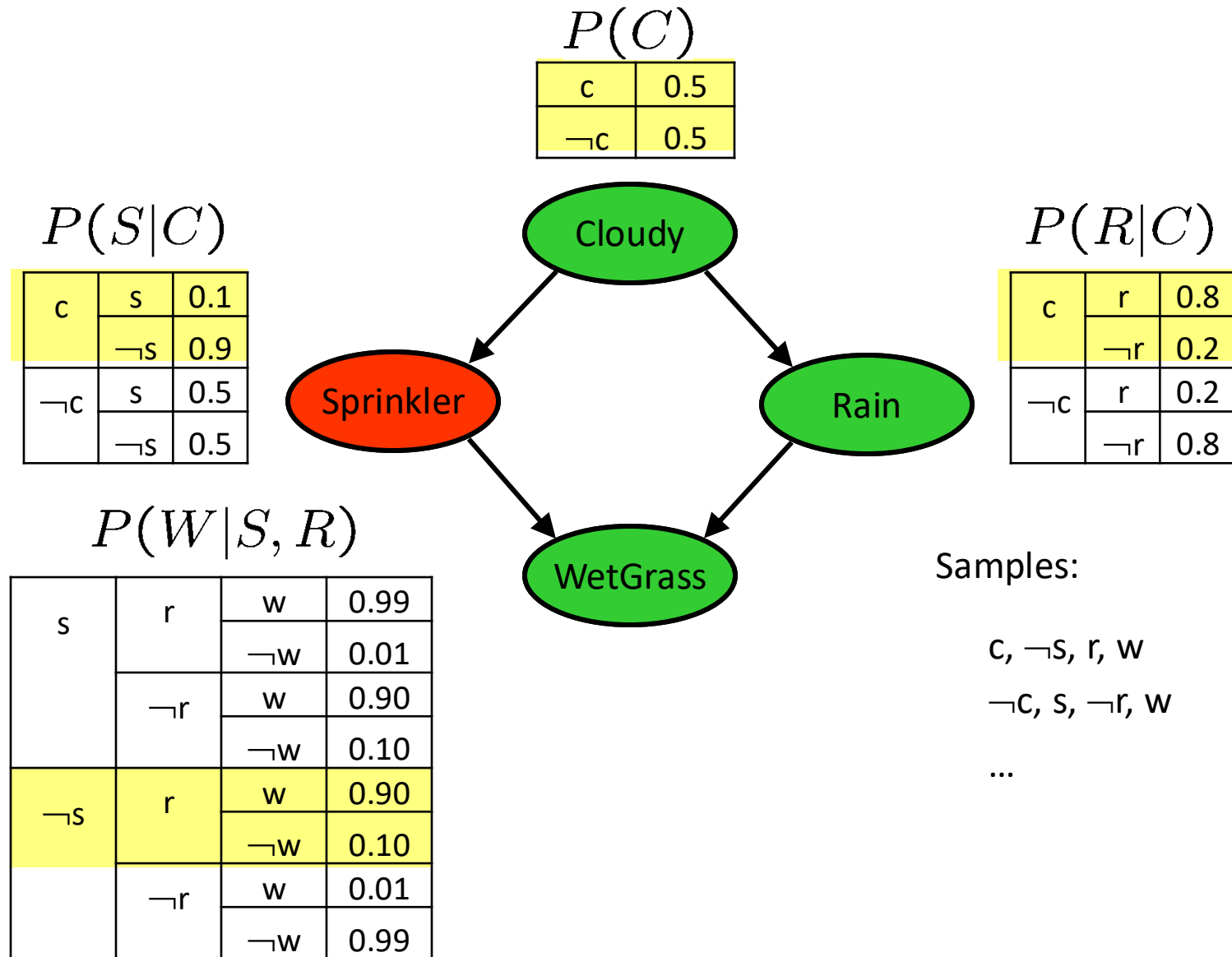
Sampling in Bayes Nets

- Prior Sampling
- Rejection Sampling
- Likelihood Weighting
- Gibbs Sampling

Prior Sampling

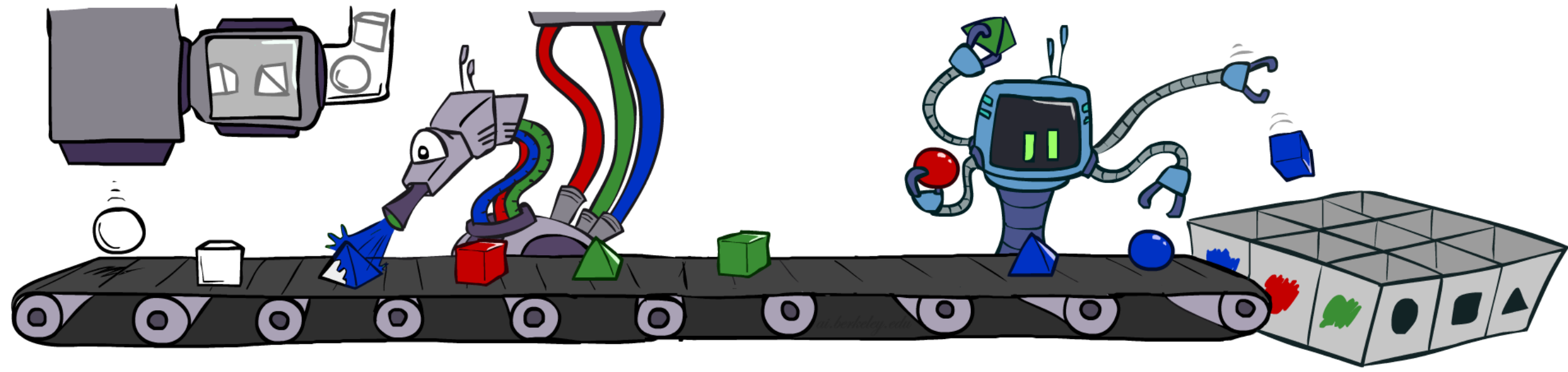


Prior Sampling



Prior Sampling

- For $i=1, 2, \dots, n$ (in topological order)
 - Sample X_i from $P(X_i \mid \text{parents}(X_i))$
- Return $(x_1, x_2, \dots, x_n)$



Using samples

- We'll get a bunch of samples from the BN:

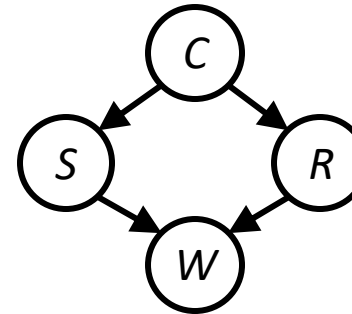
$C, \neg S, r, W$

$\neg C, S, r, W$

$\neg C, S, r, \neg W$

$C, \neg S, r, W$

$\neg C, \neg S, \neg r, W$



- If we want to know $P(W)$
 - We have counts $\langle w:4, \neg w:1 \rangle$
 - Normalize to get $P(W) = \langle w:0.8, \neg w:0.2 \rangle$
 - This will get closer to the true distribution with more samples
- If we want to know $P(C | r, w)$
 - Count (c, r, w) and $(\neg c, r, w)$
 - Normalize to get $P(C | r, w) = \langle c:0.67, \neg c:0.33 \rangle$

Prior Sampling

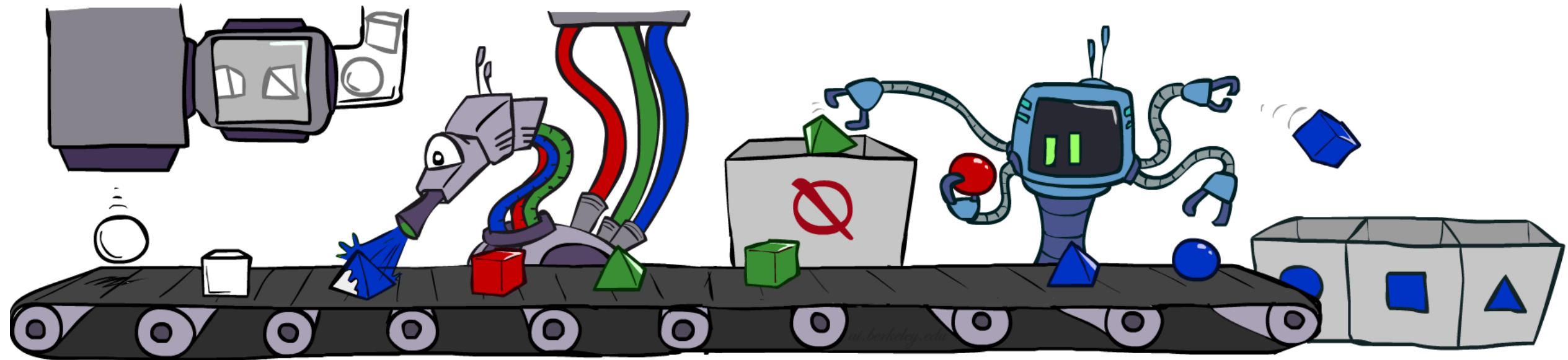
- This process generates samples with probability:

$$S_{PS}(x_1 \dots x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(X_i)) = P(x_1 \dots x_n)$$

...i.e. the BN's joint probability

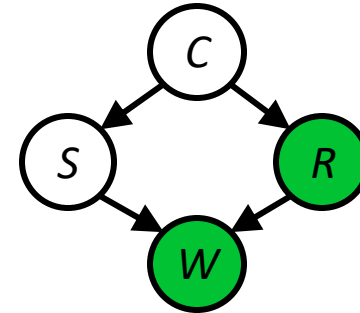
- Let the number of samples of an assignment be $N_{PS}(x_1 \dots x_n)$
- So $\hat{P}(x_1, \dots, x_n) = N_{PS}(x_1, \dots, x_n)/N$
- Then
$$\begin{aligned} \lim_{N \rightarrow \infty} \hat{P}(x_1, \dots, x_n) &= \lim_{N \rightarrow \infty} N_{PS}(x_1, \dots, x_n)/N \\ &= S_{PS}(x_1, \dots, x_n) \\ &= P(x_1 \dots x_n) \end{aligned}$$
- I.e., the sampling procedure is **consistent**

Rejection Sampling



Rejection Sampling

- A simple modification of prior sampling for conditional probabilities
- Let's say we want $P(C \mid r, w)$
- When generating a sample, reject it immediately if not $R=\text{true}$, $W=\text{true}$
- It is consistent for conditional probabilities (i.e., correct in the limit)



$C, \neg S, r, w$

~~$C, S, \neg r$~~

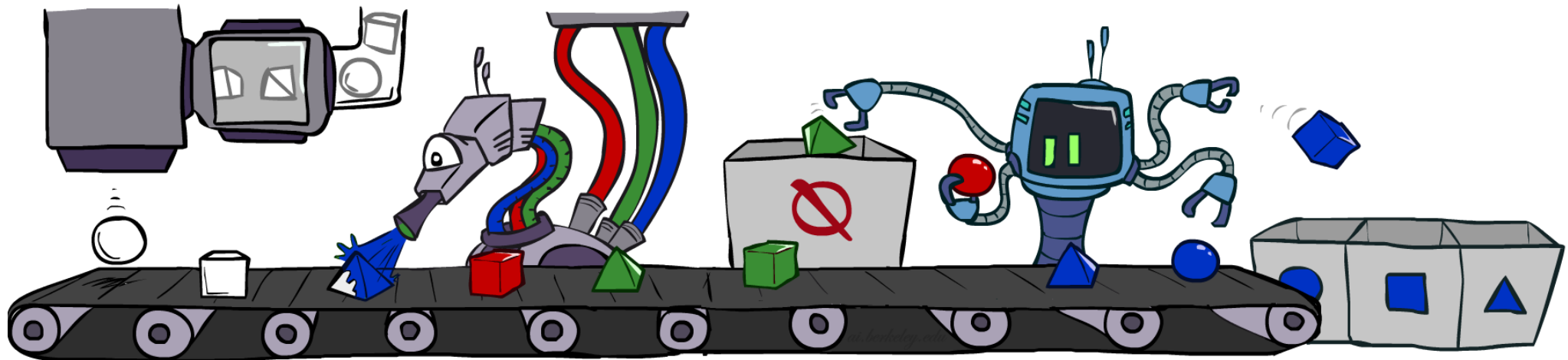
~~$\neg C, S, r, \neg w$~~

~~$C, \neg S, \neg r$~~

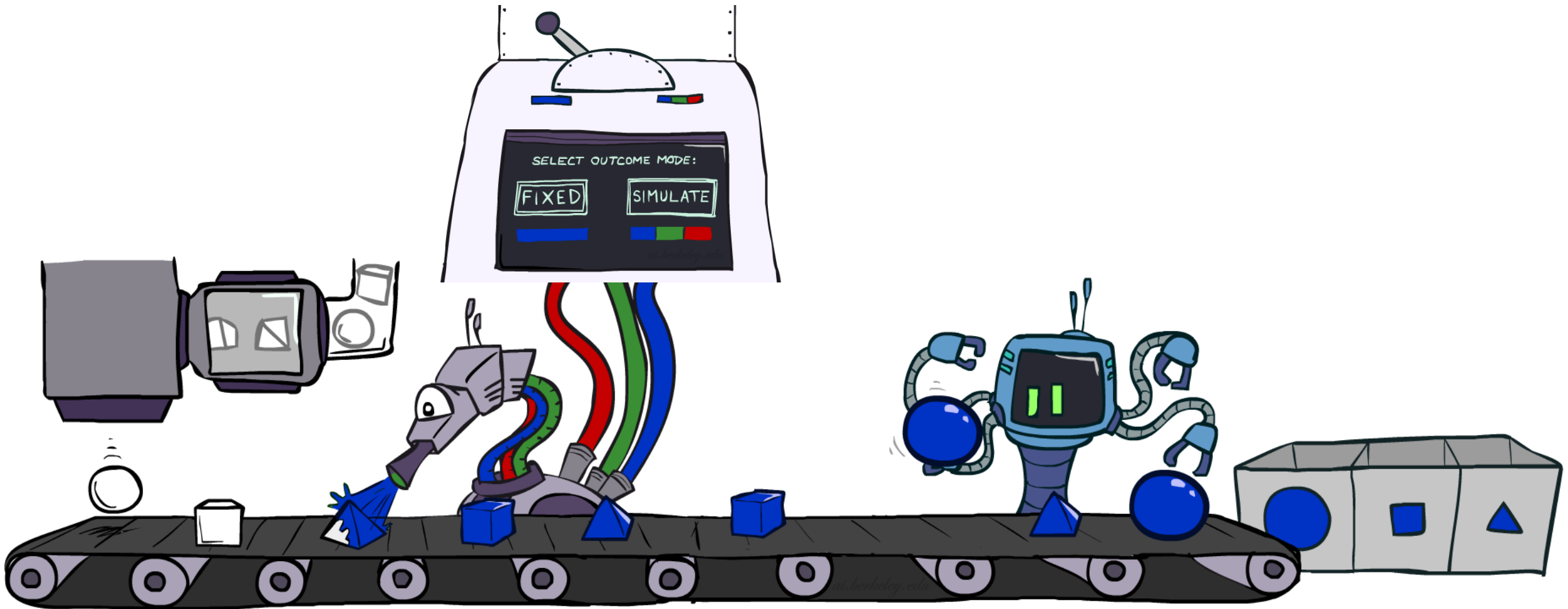
$\neg C, \neg S, r, w$

Rejection Sampling

- Input: evidence $e_1, \dots, e_k$
- For $i=1, 2, \dots, n$
 - Sample x_i from $P(x_i \mid \text{parents}(x_i))$
 - If x_i not consistent with evidence
 - Reject: Return, and no sample is generated in this cycle
- Return $(x_1, x_2, \dots, x_n)$

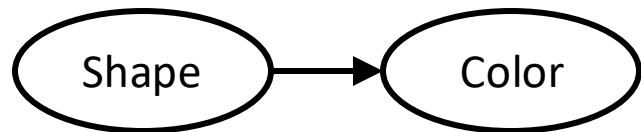


Likelihood Weighting

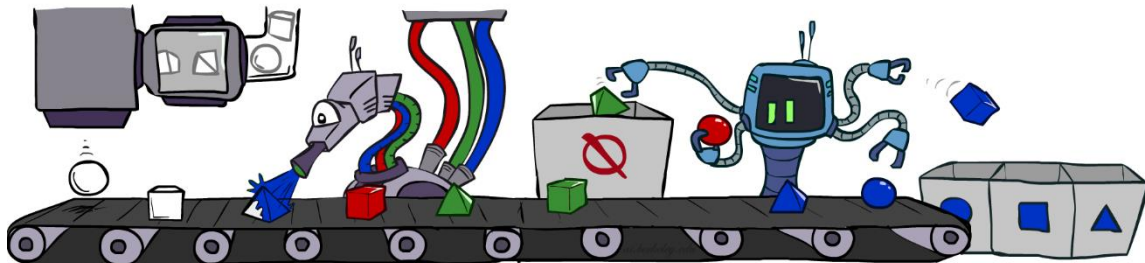


Likelihood Weighting

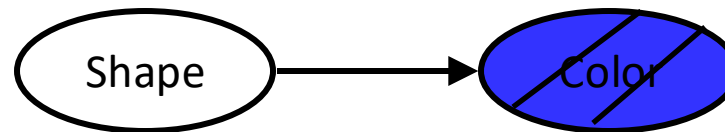
- Problem with rejection sampling:
 - If evidence is unlikely, rejects lots of samples
 - Evidence not exploited as you sample
 - Consider $P(\text{Shape} | \text{Color}=\text{blue})$



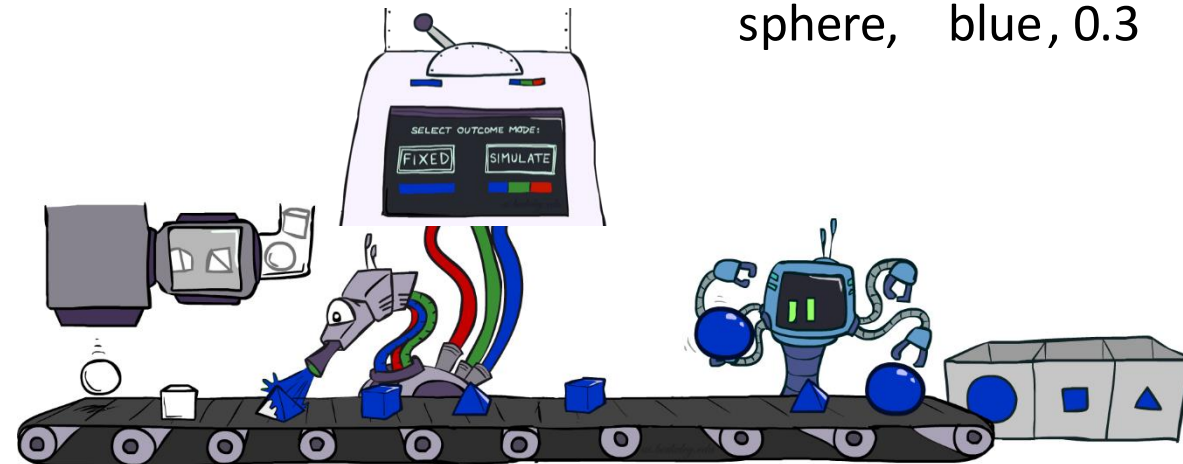
~~pyramid, green~~
~~pyramid, red~~
sphere, blue
~~cube, red~~
~~sphere, green~~



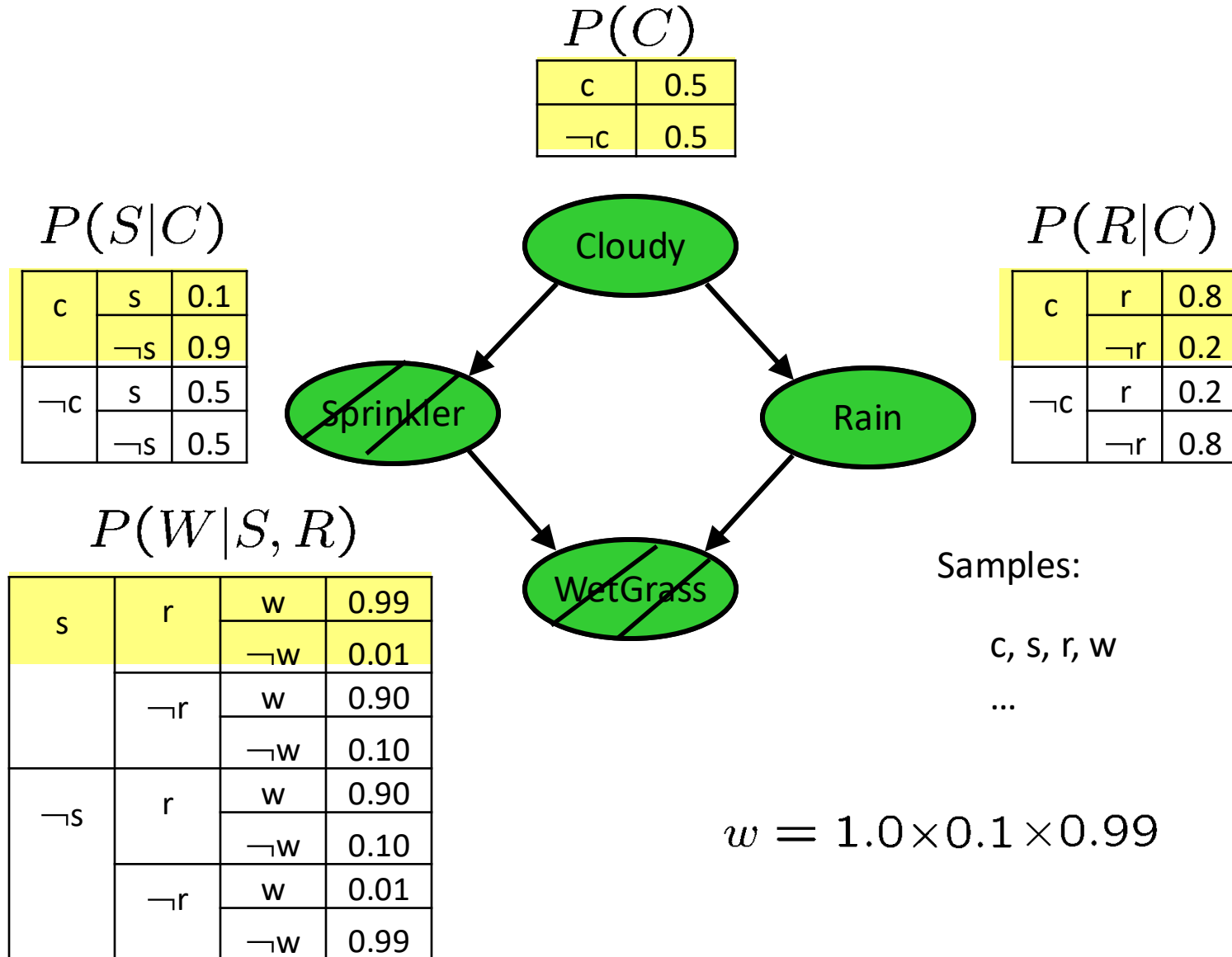
- Idea: fix evidence variables, sample the rest
 - Problem: sample distribution not consistent!
 - Solution: **weight** each sample by probability of evidence variables given parents



pyramid, blue, 0.4
pyramid, blue, 0.4
sphere, blue, 0.3
cube, blue, 0.8
sphere, blue, 0.3

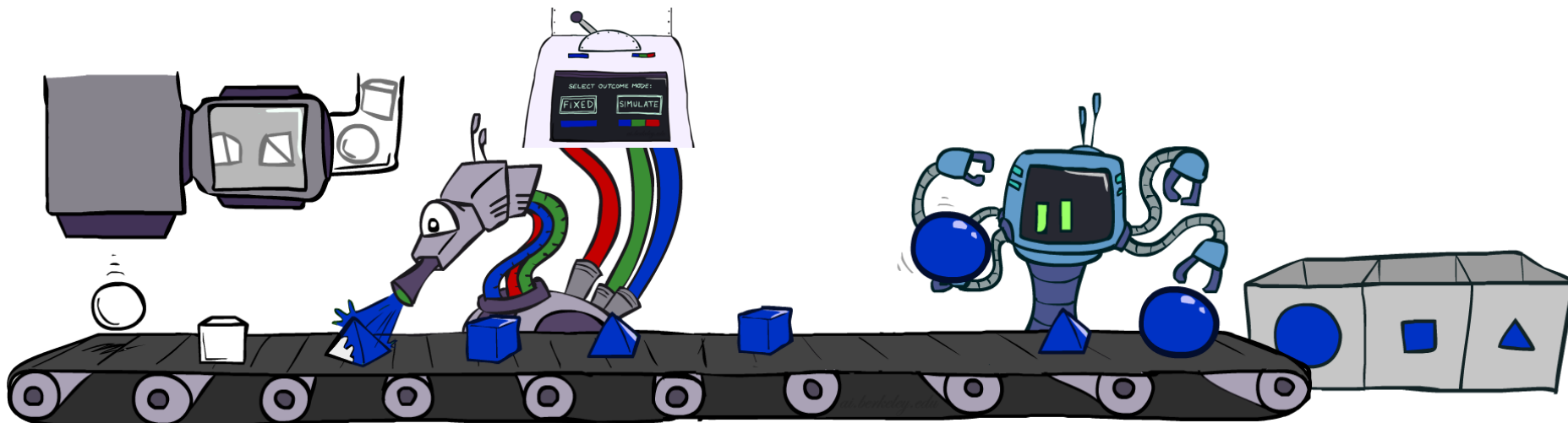


Likelihood Weighting



Likelihood Weighting

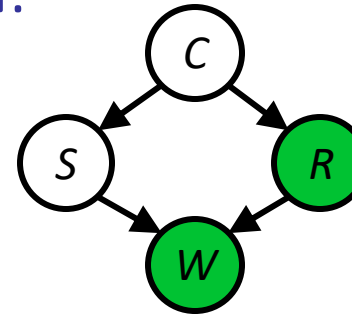
- Input: evidence $e_1, \dots, e_k$
- $w = 1.0$
- for $i=1, 2, \dots, n$
 - if X_i is an evidence variable
 - $x_i = \text{observed value}_i \text{ for } X_i$
 - Set $w = w * P(x_i \mid \text{Parents}(X_i))$
 - else
 - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
- return $(x_1, x_2, \dots, x_n), w$



Using samples

- We'll get a bunch of weighted samples from the BN:

$c, \neg s, r, w$	0.1
c, s, r, w	0.2
$\neg c, s, r, w$	0.3
$c, \neg s, r, w$	0.1
$\neg c, \neg s, r, w$	0.5



- If we want to know $P(C \mid r, w)$
 - We have weight sums $\langle (c, r, w): 0.4, (\neg c, r, w): 0.8 \rangle$
 - Normalize to get $P(C \mid r, w) = \langle c: 0.33, \neg c: 0.67 \rangle$
 - This will get closer to the true distribution with more samples

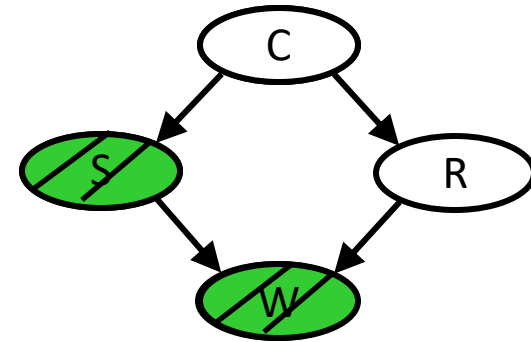
Likelihood Weighting

- Sampling distribution (z is sampled and e is fixed evidence)

$$S_{WS}(z, e) = \prod_{i=1}^l P(z_i | \text{Parents}(Z_i))$$

- Now, samples have weights

$$w(z, e) = \prod_{i=1}^m P(e_i | \text{Parents}(E_i))$$

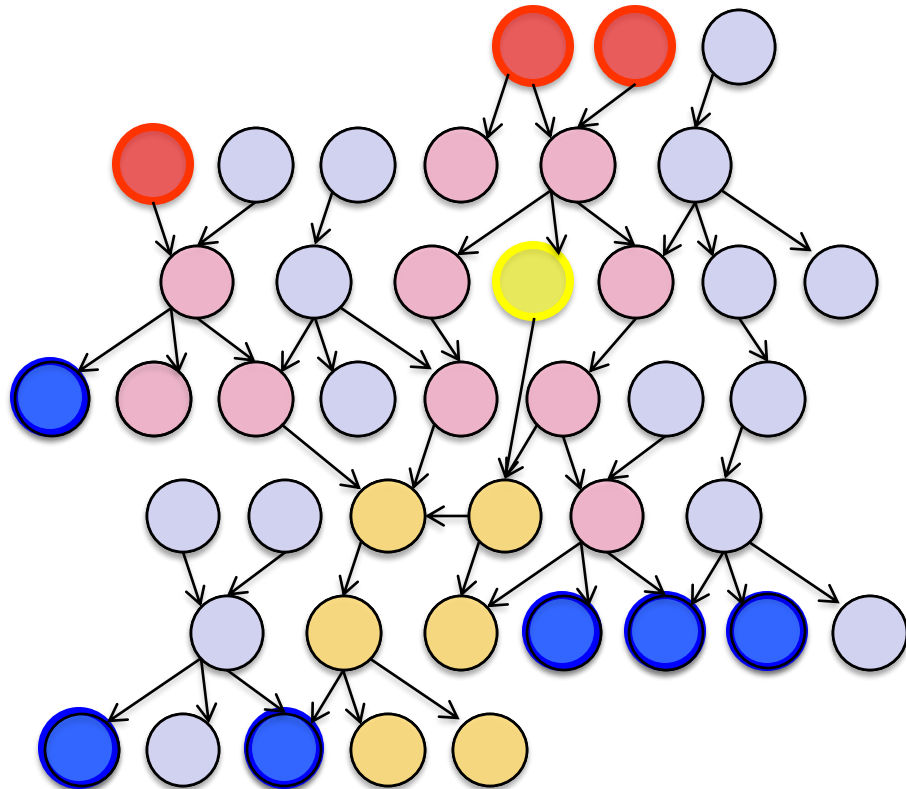


- Together, weighted sampling distribution is consistent

$$\begin{aligned} S_{WS}(z, e) \cdot w(z, e) &= \prod_{i=1}^l P(z_i | \text{Parents}(z_i)) \prod_{i=1}^m P(e_i | \text{Parents}(e_i)) \\ &= P(z, e) \end{aligned}$$

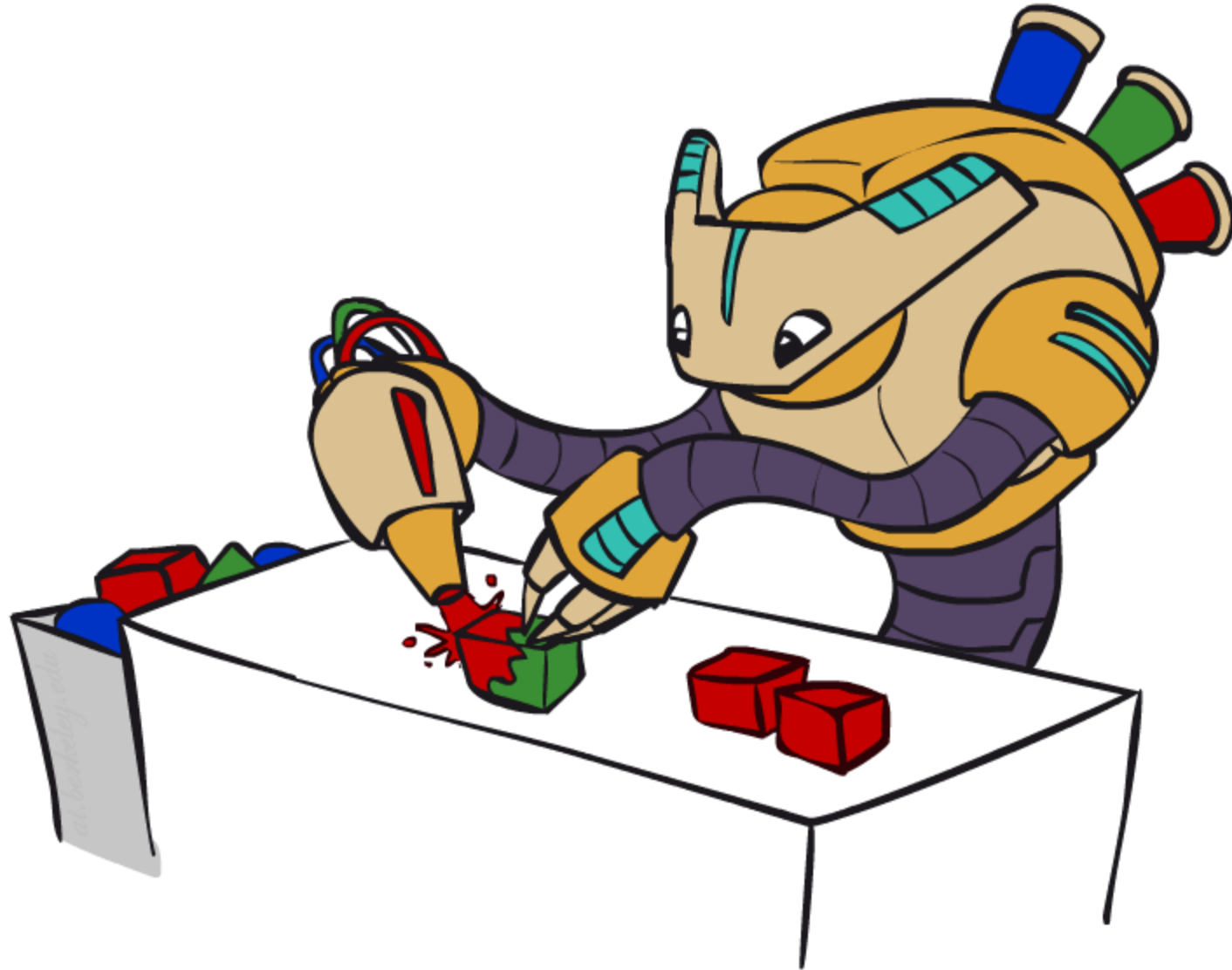
Likelihood Weighting

- Likelihood weighting is good
 - All samples are used
 - The values of *downstream* variables are influenced by *upstream* evidence



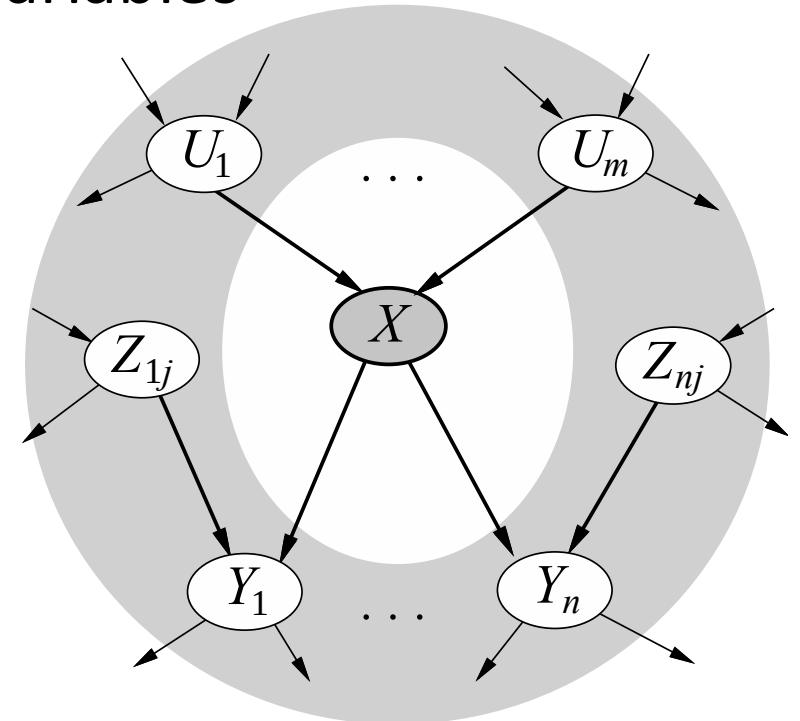
- Likelihood weighting still has weaknesses
 - The values of *upstream* variables are unaffected by *downstream* evidence
 - With many downstream evidence, we may
 - mostly get samples that are inconsistent with the evidence and thus have very small weights
 - get a few lucky samples with very large weights, which dominate the result
- We would like each variable to “see” *all* the evidence!

Gibbs Sampling



Gibbs Sampling

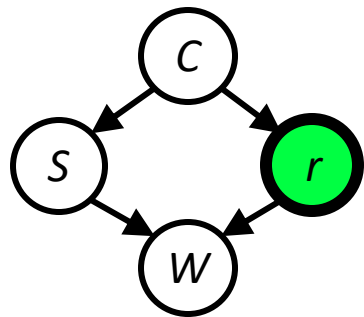
- Generate each sample by making a random change to the preceding sample
 - Evidence variables remain fixed. For each of the non-evidence variable, sample its value conditioned on all the other variables
 - $X_i' \sim P(X_i \mid x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$
 - In a Bayes net
$$P(X_i \mid x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$
$$= P(X_i \mid \text{markov_blanket}(X_i))$$
$$= \alpha P(X_i \mid u_1, \dots, u_m) \prod_j P(y_j \mid \text{parents}(Y_j))$$



Gibbs Sampling Example: $P(S | r)$

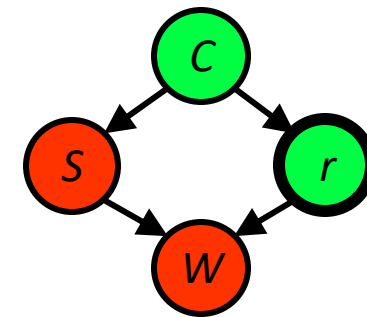
- Step 1: Fix evidence

- $R = \text{true}$



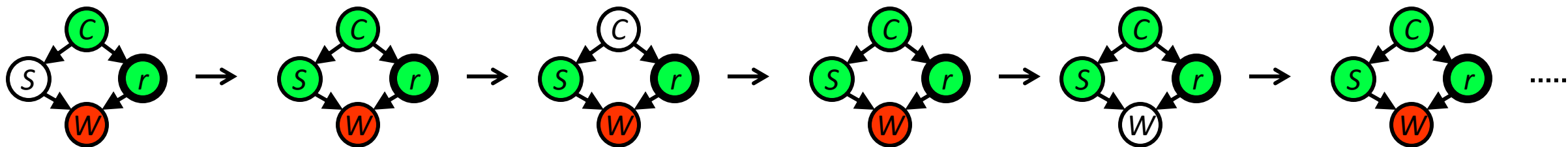
- Step 2: Initialize other variables

- Randomly



- Step 3: Repeat

- Choose an arbitrary non-evidence variable X
- Resample X from $P(X | \text{markov_blanket}(X))$

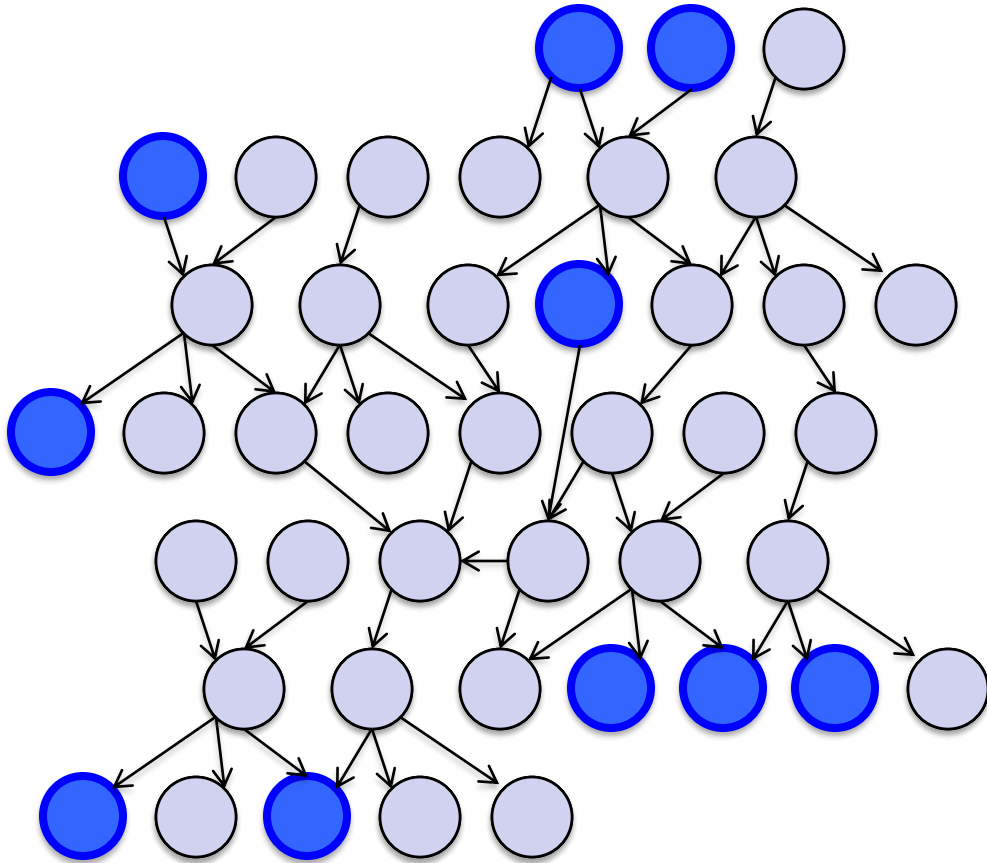


Sample $S \sim P(S | c, r, \neg w)$

Sample $C \sim P(C | s, r)$

Sample $W \sim P(W | s, r)$

Why doing this?



- Samples soon begin to reflect all the evidence in the network
- Eventually they are being drawn from the true posterior!
- Theorem: Gibbs sampling is consistent

- MCM
som

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- M



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walk”),

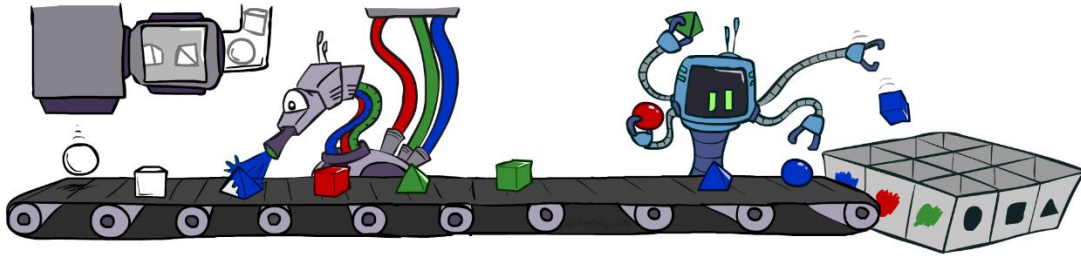
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Markov Chain Monte Carlo (MCMC)

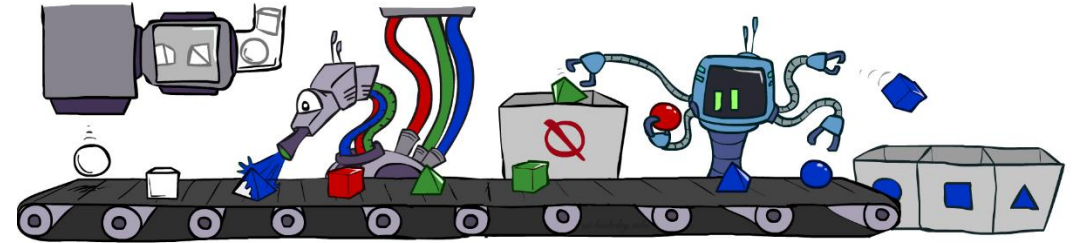
- MCMC is a family of randomized algorithms for approximating some quantity of interest over a very large state space
 - Markov chain = a sequence of randomly chosen states (“random walk”), where each state is chosen conditioned on the previous state
 - ~~Monte Carlo = a very expensive city in Monaco with a famous casino~~
 - Monte Carlo = an algorithm (usually based on sampling) that is likely to find a correct answer
- MCMC = sampling by constructing a Markov chain
- Gibbs, Metropolis-Hastings, Hamiltonian, Slice, etc.

Summary

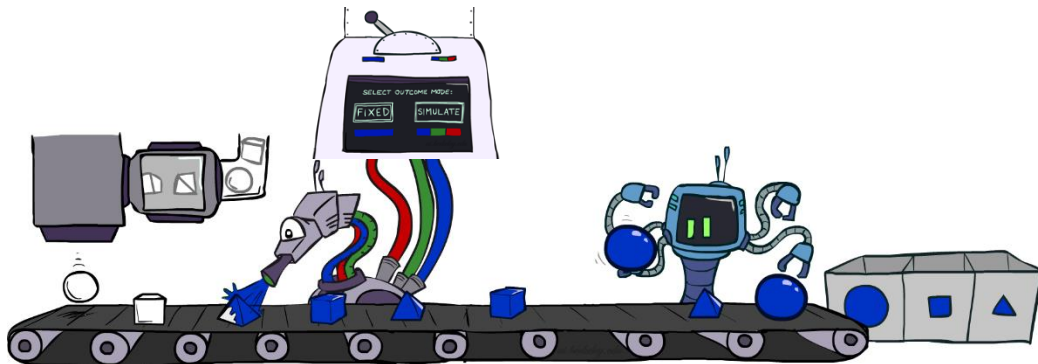
- Prior Sampling P



- Rejection Sampling $P(Q | e)$



- Likelihood Weighting $P(Q | e)$



- Gibbs Sampling $P(Q | e)$

